Effect of green tea and (-)-epigallocatechin gallate on ethanol-induced toxicity in HepG2 cells.

Despite the continuing reports supporting the hepatoprotective effects of green tea against ethanol intoxication, there remain controversies regarding the active compound(s) and molecular mechanism. These issues were addressed in the present study using cultured HepG2 cells exposed to a lethal dose of ethanol. Gamma-glutamyl transferase (GGT) was chosen as a marker of ethanol toxicity because it is widely used in clinics. When the cells were treated with ethanol at various concentrations, there was a dose-dependent increase of GGT activity in the culture media and loss of cell viability. Pretreatment of the cells with green tea extract attenuated the changes significantly. Among the green tea constituents, (-)-epigallocatechin gallate (EGCG) attenuated the ethanol cytotoxicity effectively, whereas L-theanine and caffeine had no effects. The ethanol cytotoxicity was also attenuated by alcohol dehydrogenase inhibitor 4-methyl pyrazol and GGT inhibitor acivicin as well as by thiol modulators such as S-adenosyl-L-methionine, N-acetyl-L-cysteine and glutathione. EGCG failed to prevent the intracellular glutathione loss caused by ethanol, but it appeared to be a strong GGT inhibitor. Therefore the cytoprotective effects of green tea could be attributed to the inhibition of GGT activity by EGCG. This study suggests that GGT inhibitors including EGCG may provide a novel strategy for attenuating ethanol-induced liver damage.